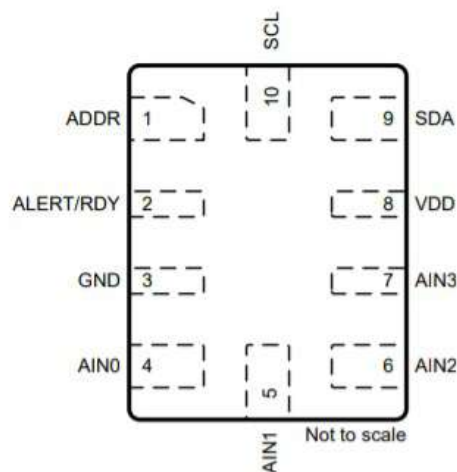


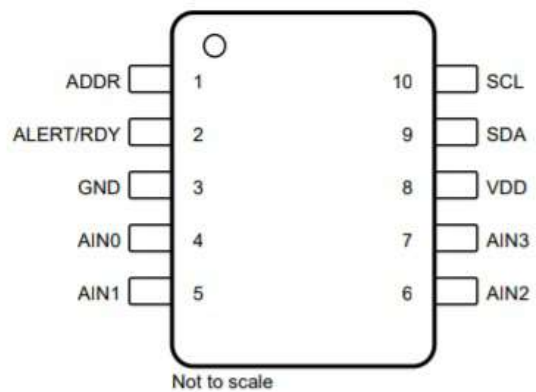
## Device Comparison Table

| DEVICE                  | RESOLUTION (Bits) | MAXIMUM SAMPLE RATE (SPS) | INPUT CHANNELS Differential (Single-Ended) | PGA | INTERFACE        | SPECIAL FEATURES   |
|-------------------------|-------------------|---------------------------|--------------------------------------------|-----|------------------|--------------------|
| <a href="#">ADS1115</a> | 16                | 860                       | 2 (4)                                      | Yes | I <sup>2</sup> C | Comparator         |
| <a href="#">ADS1114</a> | 16                | 860                       | 1 (1)                                      | Yes | I <sup>2</sup> C | Comparator         |
| <a href="#">ADS1113</a> | 16                | 860                       | 1(1)                                       | No  | I <sup>2</sup> C | None               |
| <a href="#">ADS1015</a> | 12                | 3300                      | 2 (4)                                      | Yes | I <sup>2</sup> C | Comparator         |
| <a href="#">ADS1014</a> | 12                | 3300                      | 1 (1)                                      | Yes | I <sup>2</sup> C | Comparator         |
| <a href="#">ADS1013</a> | 12                | 3300                      | 1 (1)                                      | No  | I <sup>2</sup> C | None               |
| <a href="#">ADS1118</a> | 16                | 860                       | 2 (4)                                      | Yes | SPI              | Temperature sensor |
| <a href="#">ADS1018</a> | 12                | 3300                      | 2 (4)                                      | Yes | SPI              | Temperature sensor |

## 4 Pin Configuration and Functions



**Figure 4-1. RUG Package, 10-Pin (Top View)**



**Figure 4-2. DYN and DGS Packages, 10-Pin (Top View)**

**Table 4-1. Pin Functions: RUG, DYN, and DGS Packages**

| NAME      | PIN     |         |         | TYPE           | DESCRIPTION <sup>(1)</sup>                                                                                                   |
|-----------|---------|---------|---------|----------------|------------------------------------------------------------------------------------------------------------------------------|
|           | ADS1113 | ADS1114 | ADS1115 |                |                                                                                                                              |
| ADDR      | 1       | 1       | 1       | Digital input  | I <sup>2</sup> C target address select                                                                                       |
| AIN0      | 4       | 4       | 4       | Analog input   | Analog input 0                                                                                                               |
| AIN1      | 5       | 5       | 5       | Analog input   | Analog input 1                                                                                                               |
| AIN2      | —       | —       | 6       | Analog input   | Analog input 2 (ADS1115 only)                                                                                                |
| AIN3      | —       | —       | 7       | Analog input   | Analog input 3 (ADS1115 only)                                                                                                |
| ALERT/RDY | —       | 2       | 2       | Digital output | Comparator output or conversion ready (ADS1114 and ADS1115 only). Open-drain output. Connect to VDD using a pullup resistor. |
| GND       | 3       | 3       | 3       | Analog         | Ground                                                                                                                       |
| NC        | 2, 6, 7 | 6, 7    | —       | —              | No connect. Leave the pin floating or connect to GND.                                                                        |
| SCL       | 10      | 10      | 10      | Digital input  | Serial clock input. Connect to VDD using a pullup resistor.                                                                  |
| SDA       | 9       | 9       | 9       | Digital I/O    | Serial data input and output. Connect to VDD using a pullup resistor.                                                        |
| VDD       | 8       | 8       | 8       | Analog         | Power supply. Connect a 0.1μF, power-supply decoupling capacitor to GND.                                                     |

(1) See the [Unused Inputs and Outputs](#) section for unused pin connections.

## 5 Specifications

### 5.1 Absolute Maximum Ratings

over operating ambient temperature range (unless otherwise noted)<sup>(1)</sup>

|                           |                                   | MIN       | MAX       | UNIT |
|---------------------------|-----------------------------------|-----------|-----------|------|
| Power-supply voltage      | VDD to GND                        | −0.3      | 7         | V    |
| Analog input voltage      | AIN0, AIN1, AIN2, AIN3            | GND − 0.3 | VDD + 0.3 | V    |
| Digital input voltage     | SDA, SCL, ADDR, ALERT/RDY         | GND − 0.3 | 5.5       | V    |
| Input current, continuous | Any pin except power supply pins  | −10       | 10        | mA   |
| Temperature               | Operating ambient, T <sub>A</sub> | −40       | 125       | °C   |
|                           | Junction, T <sub>J</sub>          | −40       | 150       |      |
|                           | Storage, T <sub>stg</sub>         | −60       | 150       |      |

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

### 5.2 ESD Ratings

|                    |                         |                                                                                | VALUE | UNIT |
|--------------------|-------------------------|--------------------------------------------------------------------------------|-------|------|
| V <sub>(ESD)</sub> | Electrostatic discharge | Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 <sup>(1)</sup>              | ±2000 | V    |
|                    |                         | Charged-device model (CDM), per JEDEC specification JESD22-C101 <sup>(2)</sup> | ±500  |      |

- (1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.

- (2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

### 5.3 Recommended Operating Conditions

|                                    |                                                                                                              | MIN    | NOM | MAX    | UNIT |
|------------------------------------|--------------------------------------------------------------------------------------------------------------|--------|-----|--------|------|
| <b>POWER SUPPLY</b>                |                                                                                                              |        |     |        |      |
|                                    | Power supply (VDD to GND)                                                                                    | 2      |     | 5.5    | V    |
| <b>ANALOG INPUTS<sup>(1)</sup></b> |                                                                                                              |        |     |        |      |
| FSR                                | Full-scale input voltage range <sup>(2)</sup> (V <sub>IN</sub> = V <sub>(AINP)</sub> − V <sub>(AINN)</sub> ) | ±0.256 |     | ±6.144 | V    |
| V <sub>(AINx)</sub>                | Absolute input voltage                                                                                       | GND    |     | VDD    | V    |
| <b>DIGITAL INPUTS</b>              |                                                                                                              |        |     |        |      |
| V <sub>DIG</sub>                   | Digital input voltage                                                                                        | GND    |     | 5.5    | V    |
| <b>TEMPERATURE</b>                 |                                                                                                              |        |     |        |      |
| T <sub>A</sub>                     | Operating ambient temperature                                                                                | −40    |     | 125    | °C   |

- (1) AINP and AINN denote the selected positive and negative inputs. AINx denotes one of the four available analog inputs.

- (2) This parameter expresses the full-scale range of the ADC scaling. No more than VDD + 0.3V must be applied to the analog inputs of the device. See [Table 7-1](#) for more information.

### 5.4 Thermal Information

| THERMAL METRIC <sup>(1)</sup> |                                              | RUG (X2QFN) | DYN (SOT) | DGS (VSSOP) | UNIT |
|-------------------------------|----------------------------------------------|-------------|-----------|-------------|------|
|                               |                                              | 10 PINS     | 10 PINS   | 10 PINS     |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance       | 245.2       | 147.1     | 182.7       | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance    | 69.3        | 59.3      | 67.2        | °C/W |
| R <sub>θJB</sub>              | Junction-to-board thermal resistance         | 172.0       | 71.3      | 103.8       | °C/W |
| ψ <sub>JT</sub>               | Junction-to-top characterization parameter   | 8.2         | 2.8       | 10.2        | °C/W |
| ψ <sub>JB</sub>               | Junction-to-board characterization parameter | 170.8       | 70.4      | 102.1       | °C/W |
| R <sub>θJC(bot)</sub>         | Junction-to-case (bottom) thermal resistance | N/A         | N/A       | N/A         | °C/W |

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application note.

## 5.5 Electrical Characteristics

at VDD = 3.3V, data rate = 8SPS, and full-scale input voltage range (FSR) =  $\pm 2.048\text{V}$  (unless otherwise noted); maximum and minimum specifications apply from T<sub>A</sub> = -40°C to +125°C; typical specifications are at T<sub>A</sub> = 25°C

| PARAMETER            |                                            | TEST CONDITIONS                                                                 | MIN                                           | TYP     | MAX        | UNIT                    |
|----------------------|--------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------|---------|------------|-------------------------|
| ANALOG INPUT         |                                            |                                                                                 |                                               |         |            |                         |
|                      | Common-mode input impedance                | FSR = $\pm 6.144\text{V}^{(1)}$                                                 | 10                                            |         | M $\Omega$ |                         |
|                      |                                            | FSR = $\pm 4.096\text{V}^{(1)}$ , FSR = $\pm 2.048\text{V}$                     | 6                                             |         |            |                         |
|                      |                                            | FSR = $\pm 1.024\text{V}$                                                       | 3                                             |         |            |                         |
|                      |                                            | FSR = $\pm 0.512\text{V}$ , FSR = $\pm 0.256\text{V}$                           | 100                                           |         |            |                         |
|                      | Differential input impedance               | FSR = $\pm 6.144\text{V}^{(1)}$                                                 | 22                                            |         | M $\Omega$ |                         |
|                      |                                            | FSR = $\pm 4.096\text{V}^{(1)}$                                                 | 15                                            |         |            |                         |
|                      |                                            | FSR = $\pm 2.048\text{V}$                                                       | 4.9                                           |         |            |                         |
|                      |                                            | FSR = $\pm 1.024\text{V}$                                                       | 2.4                                           |         |            |                         |
|                      |                                            | FSR = $\pm 0.512\text{V}$ , $\pm 0.256\text{V}$                                 | 710                                           |         | k $\Omega$ |                         |
| SYSTEM PERFORMANCE   |                                            |                                                                                 |                                               |         |            |                         |
|                      | Resolution (no missing codes)              |                                                                                 | 16                                            |         |            | Bits                    |
| DR                   | Data rate                                  |                                                                                 | 8, 16, 32, 64, 128, 250, 475, 860             |         |            | SPS                     |
|                      | Data rate variation                        | All data rates                                                                  | -10%10%                                       |         |            |                         |
|                      | Output noise                               |                                                                                 | See <a href="#">Noise Performance</a> section |         |            |                         |
| INL                  | Integral nonlinearity                      | DR = 8SPS, FSR = $\pm 2.048\text{V}^{(2)}$                                      | 1                                             |         |            | LSB                     |
|                      | Offset error                               | FSR = $\pm 2.048\text{V}$ , differential inputs                                 | -3                                            | $\pm 1$ | 3          | LSB                     |
|                      |                                            | FSR = $\pm 2.048\text{V}$ , single-ended inputs                                 |                                               | $\pm 3$ |            |                         |
|                      | Offset drift over temperature              | FSR = $\pm 2.048\text{V}$                                                       | 0.005                                         |         |            | LSB/ $^{\circ}\text{C}$ |
|                      | Long-term Offset drift                     | FSR = $\pm 2.048\text{V}$ , $T_{\text{A}}$ = $125^{\circ}\text{C}$ , 1000 hours | $\pm 1$                                       |         |            | LSB                     |
|                      | Offset power-supply rejection              | FSR = $\pm 2.048\text{V}$ , DC supply variation                                 | 1                                             |         |            | LSB/V                   |
|                      | Offset channel match                       | Match between any two inputs                                                    | 3                                             |         |            | LSB                     |
|                      | Gain error <sup>(3)</sup>                  | FSR = $\pm 2.048\text{V}$ , $T_{\text{A}}$ = $25^{\circ}\text{C}$               | 0.01%0.15%                                    |         |            |                         |
|                      | Gain drift over temperature <sup>(3)</sup> | FSR = $\pm 0.256\text{V}$                                                       | 7                                             |         | 40         | ppm/ $^{\circ}\text{C}$ |
|                      |                                            | FSR = $\pm 2.048\text{V}$                                                       | 5                                             |         |            |                         |
|                      |                                            | FSR = $\pm 6.144\text{V}^{(1)}$                                                 | 5                                             |         |            |                         |
|                      | Long-term gain drift <sup>(3)</sup>        | FSR = $\pm 2.048\text{V}$ , $T_{\text{A}}$ = $125^{\circ}\text{C}$ , 1000 hours | $\pm 0.05\%$                                  |         |            |                         |
|                      | Gain power-supply rejection                |                                                                                 | 80                                            |         |            | ppm/V                   |
|                      | Gain match <sup>(3)</sup>                  | Match between any two gains                                                     | 0.02%0.1%                                     |         |            |                         |
|                      | Gain channel match                         | Match between any two inputs                                                    | 0.05%0.1%                                     |         |            |                         |
| CMRR                 | Common-mode rejection ratio                | At DC, FSR = $\pm 0.256\text{V}$                                                | 105                                           |         | dB         |                         |
|                      |                                            | At DC, FSR = $\pm 2.048\text{V}$                                                | 100                                           |         |            |                         |
|                      |                                            | At DC, FSR = $\pm 6.144\text{V}^{(1)}$                                          | 90                                            |         |            |                         |
|                      |                                            | $f_{\text{CM}}$ = 60Hz, DR = 8SPS                                               | 105                                           |         |            |                         |
|                      |                                            | $f_{\text{CM}}$ = 50Hz, DR = 8SPS                                               | 105                                           |         |            |                         |
| DIGITAL INPUT/OUTPUT |                                            |                                                                                 |                                               |         |            |                         |
| V <sub>IH</sub>      | High-level input voltage                   |                                                                                 | 0.7 VDD                                       |         | 5.5        | V                       |
| V <sub>IL</sub>      | Low-level input voltage                    |                                                                                 | GND                                           |         | 0.3 VDD    | V                       |
| V <sub>OL</sub>      | Low-level output voltage                   | I <sub>OL</sub> = 3mA                                                           | GND                                           |         | 0.150.4    | V                       |
|                      | Input leakage current                      | GND < V <sub>DIG</sub> < VDD                                                    | -10                                           |         | 10         | $\mu\text{A}$           |

## 5.5 Electrical Characteristics (continued)

at VDD = 3.3V, data rate = 8SPS, and full-scale input voltage range (FSR) = ±2.048V (unless otherwise noted); maximum and minimum specifications apply from TA = –40°C to +125°C; typical specifications are at TA = 25°C

| PARAMETER           | TEST CONDITIONS   | MIN         | TYP       | MAX | UNIT |
|---------------------|-------------------|-------------|-----------|-----|------|
| <b>POWER-SUPPLY</b> |                   |             |           |     |      |
| I <sub>VDD</sub>    | Supply current    | Power-down  | TA = 25°C | 0.5 | 2    |
|                     |                   |             |           |     | 5    |
|                     | Operating         | TA = 25°C   |           | 150 | 200  |
|                     |                   |             |           |     | 300  |
| P <sub>D</sub>      | Power dissipation | VDD = 5.0V  |           | 0.9 | mW   |
|                     |                   | VDD = 3.3 V |           | 0.5 |      |
|                     |                   | VDD = 2.0V  |           | 0.3 |      |

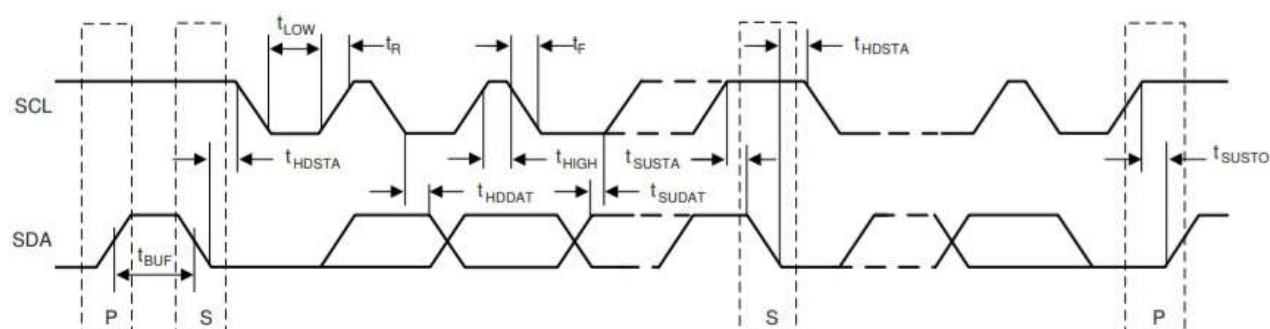
- (1) This parameter expresses the full-scale range of the ADC scaling. No more than VDD + 0.3V must be applied to the analog inputs of the device. See Table 7-1 for more information.
- (2) Best-fit INL; covers 99% of full-scale
- (3) Includes all errors from onboard PGA and voltage reference

## 5.6 Timing Requirements: I<sup>2</sup>C

over operating ambient temperature range and VDD = 2.0V to 5.5V (unless otherwise noted)

|                    |                                                                                            | FAST MODE |     | HIGH-SPEED MODE |     | UNIT |
|--------------------|--------------------------------------------------------------------------------------------|-----------|-----|-----------------|-----|------|
|                    |                                                                                            | MIN       | MAX | MIN             | MAX |      |
| f <sub>SCL</sub>   | SCL clock frequency                                                                        | 0.01      | 0.4 | 0.01            | 3.4 | MHz  |
| t <sub>BUF</sub>   | Bus free time between START and STOP condition                                             | 600       |     | 160             |     | ns   |
| t <sub>HDSTA</sub> | Hold time after repeated START condition. After this period, the first clock is generated. | 600       |     | 160             |     | ns   |
| t <sub>SUSTA</sub> | Setup time for a repeated START condition                                                  | 600       |     | 160             |     | ns   |
| t <sub>SUSTO</sub> | Setup time for STOP condition                                                              | 600       |     | 160             |     | ns   |
| t <sub>HDDAT</sub> | Data hold time                                                                             | 0         |     | 0               |     | ns   |
| t <sub>SUDAT</sub> | Data setup time                                                                            | 100       |     | 10              |     | ns   |
| t <sub>LOW</sub>   | Low period of the SCL clock pin                                                            | 1300      |     | 160             |     | ns   |
| t <sub>HIGH</sub>  | High period for the SCL clock pin                                                          | 600       |     | 60              |     | ns   |
| t <sub>F</sub>     | Fall time for both SDA and SCL signals <sup>(1)</sup>                                      |           | 300 |                 | 160 | ns   |
| t <sub>R</sub>     | Rise time for both SDA and SCL signals <sup>(1)</sup>                                      |           | 300 |                 | 160 | ns   |

- (1) For high-speed mode maximum values, the capacitive load on the bus line must not exceed 400pF.



**Figure 5-1. I<sup>2</sup>C Interface Timing**